

Lab Manual

Important Note:

- Have an understanding of

- ✓ Friend class
- ✓ Friend function
- ✓ Association

Task 1

Create a class stat that has two data members.

- Int size
- Int *array;

Create the following member functions:

- Default constructor
- Getter
- Setter
- Friend function median(stat obj)

Your task is to find the median and return it.

Task 2

Implement two classes

- MyClass
- FriendClass.

The MyClass class has a private member secretData, which is a string representing some confidential data.

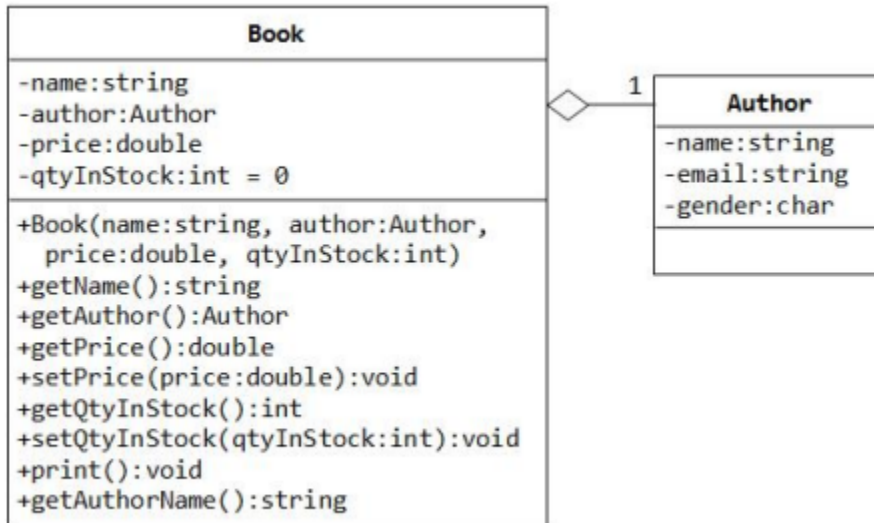
The FriendClass class has two member functions:

- displaySecretData(): The displaySecretData() function can access and display the secretData member of MyClass.
- modifySecretData(); The modifySecretData() function can modify the secretData member of MyClass as well as display the modified data.

```
int main()
{
    MyClass obj("Hello, friend!");
    FriendClass friendObj;
    friendObj.displaySecretData(obj);
    friendObj.modifySecretData(obj, "Updated secret data");
    return 0;
}
```

Task 3

Understand the diagram and write a code for it.



Implement all functions in book class

Your Author class must have the following functions

- Author(const string & name, const string & email, char gender);
- string getName() const;
- string getEmail() const;
- void setEmail(const string & email);
- char getGender() const;
- void print() const;

Your driver function must be like this.

```
int main() {
// Declare and construct an instance of Author
Author peter("Peter Jones", "peter@somewhere.com", 'm');
peter.print(); // Peter Jones (m) at peter@somewhere.com
// Declare and construct an instance of Book
Book book("C++ for Dummies", &peter, 19.99);
book.print();
// 'C++ for Dummies' by Peter Jones (m) at peter@somewhere.com
peter.setEmail("peter@xyz.com");
peter.print(); // Peter Jones (m) at peter@xyz.com
book.print();
// 'C++ for Dummies' by Peter Jones (m) at peter@somewhere.com
book.getAuthor().setEmail("peter@abc.com");
```

```
book.print();  
// 'C++ for Dummies' by Peter Jones (m) at peter@somewhere.com  
}
```

Task 4

Implement two classes:

- Student
- Course.

There is an association between them, where a Course can have multiple Student objects enrolled in it.

Class student has two data members.

- string name
- int studentId

And has following member functions.

- Student(const string& name, int studentId) : name(name), studentId(studentId);
- string getName() const;
- int getStudentId() const

Class Course has following data members

- string courseName
- Student* students

And has following members functions.

- Course(const std::string& courseName) : courseName(courseName)
- void addStudent(Student* student)
- void displayStudents() const

```
int main()  
{  
Student student1("John Smith", 12345);  
Student student2("Jane Doe", 54321);  
Course mathCourse("Mathematics");  
mathCourse.addStudent(&student1);  
mathCourse.addStudent(&student2);  
mathCourse.displayStudents();  
return 0;  
}
```